MRIDUL VERMA R

AI & Devops

Gmail — LinkedIn — GitHub — LeetCode

Summary

Final-year B.Tech student skilled in Java and Python, with experience in building scalable, maintainable software using OOP principles. Proficient in debugging, performance optimization, and clean code practices, with strong collaboration and communication skills. Passionate about solving real-world problems through efficient software development.

Technical Skills

Programming Languages: Python, Java, JavaScript (ES6+), SQL, C++
Cloud & DevOps: AWS (EC2, S3, ALB, IAM), Docker, CI/CD (CodePipeline, CodeDeploy), Git
AI & Machine Learning: Scikit-learn, CNN, Transfer Learning, Model Training, Inference Pipelines
Data Analytics: SQL (MySQL), Pandas, NumPy, Data Cleaning, EDA, Matplotlib, Seaborn
Web Technologies: React.js, Node.js, Express.js, RESTful APIs, Tailwind CSS, MongoDB

Projects

Farmer AI Assistant – AI-Based Crop Disease Detection & Advisory System Sep 2025 – Nov 2025
GitHub

- Developed a CNN-based image classification system for crop disease detection with optimized preprocessing and augmentation.
- Built a multilingual voice-enabled assistant (Tamil, Hindi, English) for accessible farmer interaction.
- Improved model performance by analyzing predictions and reducing false positives.
- Integrated real-time weather and market data to deliver context-aware crop recommendations.
- Designed an end-to-end pipeline from image input to actionable insights.

Tech Stack: Python, CNN, REST APIs

Campus Navigation Facility Management System Sep 2026 - Oct 2026
GitHub

- Implemented Dijkstra's algorithm over a custom spatial graph, achieving ¡50ms route latency.
- Designed MongoDB spatial schemas modeling cross-building waypoints and multi-floor transitions.
- Built a facility booking engine with automated collision detection to eliminate double-booking.
- Developed a responsive React UI rendering interactive floorplans and turn-by-turn directions.

Tech Stack: React.js, Node.js, Express.js, MongoDB, Spatial Graph / GIS, Tailwind CSS, RESTful APIs

Food Delivery System – Scalable Application Deployment & Infrastructure Design Mar 2026 – Present
GitHub

- Deployed a full-stack application focusing on scalability, reliability, and performance.
- Designed containerized services (Docker) for consistent deployment across environments.
- Implemented CI/CD pipelines to automate build, testing, and deployment workflows.
- Architected system for load balancing, fault tolerance, and secure access control.

Tech Stack: Docker, CI/CD, Cloud Infrastructure, Node.js

Experience

AI/ML Developer Intern Oct 2025 – Nov 2025
ShadowFox Erode, India

- Completed end-to-end ML workflows: data collection, preprocessing, model development, evaluation, and production deployment
- Gained hands-on experience transitioning ML models from experimentation to production

Education

B.E. Computer Science & Engineering Aug 2023 – May 2027
Bannari Amman Institute of Technology
CGPA: 8.52

HSLC Aug 2022 – Mar 2023
Carmel Matric Higher Secondary School
Grade: 96.8

Achievements & Certifications

- Oracle AI Foundations Associate (2025)
- Python for Data Science Issued by IBM
- Reliance Foundation Undergraduate Scholar 2023